Chapter 3 Exercises: Selected Solutions

1. (a)

$$\begin{aligned} P(A \cup B) &= 0.2 + 0.5 - 0.2 \times 0.5 \\ &= 0.6 \end{aligned}$$

(b)

$$P(A \cap B) = 0.1.$$

(c) Since

$$A^c \cap B^c = (A \cup B)^c$$

we find $P(A^c \cap B^c)$ is 0.4.

2. As there are 10 out of 38 students who received a B grade, one can assume

$$P(B) = \frac{10}{38} = 0.263$$

to 3 d.p.; now to find the "expected" value of the gain, we calculate

$$\frac{5}{38} \times 10 + \frac{10}{38} \times 5 + \frac{17}{38} \times 2 + \frac{6}{38} \times 0 = \frac{67}{19}$$

dollars, i.e., \$3.53.

3. (a)

$$\begin{aligned} P(C|+ve) &= \frac{P(+ve|C)\,P(C)}{P(+ve|C)\,P(C) + P(+ve|C^c)\,P(C^c)} \\ &= \frac{0.9 \times 0.0001}{0.9 \times 0.0001 + 0.001 \times 0.9999} \\ &= 0.00826. \end{aligned}$$

4. (a) Summing the three mutually exclusive events,

$$\begin{aligned} P(\geq 1 \text{ accident}) &= P(\geq 1 \text{ acc.} \cap \text{good}) + P(\geq 1 \text{ acc.} \cap \text{medium}) + P(\geq 1 \text{ acc.} \cap \text{poor}) \\ &= P(\geq 1 \text{ acc.}|\text{good})\,P(\text{good}) + P(\geq 1 \text{ acc.}|\text{medium})\,P(\text{medium}) \\ &\quad + P(\geq 1 \text{ acc.}|\text{poor})\,P(\text{poor}) \\ &= 0.1 \times 0.3 + 0.3 \times 0.5 + 0.5 \times 0.2 \\ &= 0.28. \end{aligned}$$

(b) Taking "one accident" to mean "at least one accident", we have

$$\begin{aligned} P\left(\text{good} \mid \geq 1 \text{ accident}\right) &= \frac{P\left(\text{good} \cap \geq 1 \text{ accident}\right)}{P\left(\geq 1 \text{ accident}\right)} \\ &= \frac{0.03}{0.28} \\ &= 0.107. \end{aligned}$$

5. xxx

6. (a)

$$\begin{aligned} P\left(EL\right) &= P\left(EL|D\right)P\left(D\right) + P\left(EL|D^c\right)P\left(D^c\right) \\ &= 0.6 \times 0.7 + 0.2 \times 0.3 \\ &= 0.48. \end{aligned}$$

(b)

$$\begin{aligned} P\left(D|EL\right) &= \frac{0.6 \times 0.7}{0.6 \times 0.7 + 0.2 \times 0.3} \\ &= 0.875. \end{aligned}$$

7. Denoting failure by the event F we have, by Bayes' Theorem,

$$\begin{aligned} P\left(A|F\right) &= \frac{P\left(F|A\right)P\left(A\right)}{P\left(F|A\right)P\left(A\right) + P\left(F|B\right)P\left(B\right)} \\ &= \frac{0.2 \times 0.7}{0.2 \times 0.7 + 0.1 \times 0.3} \\ &= 0.8235. \end{aligned}$$

8. (a) No, since 4 and 6 occur in both. (b) $A_1 \cap A_2 = \{4, 6\}$, so

$$P\left(A_1 \cap A_2\right) = \frac{1}{5}.$$

9. The sample space here consists of eight events

$$\begin{array}{cc} \{HHH\} & \{TTH\} \\ \{HHT\} & \{THT\} \\ \{HTH\} & \{HTT\} \\ \{THH\} & \{TTT\} \end{array}$$

As we are told on a single toss that

$$P(H) = 2P(T)$$

then since $P(H) + P(T) = 1$ it follows that

$$\begin{aligned} P(T) &= \frac{1}{3}, \\ P(H) &= \frac{2}{3}. \end{aligned}$$

Hence, in three tosses,

$$\begin{aligned} P(\text{Two T}) &= P(TTH) + P(THT) + P(HTT) \\ &= 3 \times \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} \\ &= \frac{2}{9}. \end{aligned}$$

10. xxx

11. If

$$A_i = \{\text{Component } i \text{ fails}\}$$

then

$$\begin{aligned} A &= \{\text{System fails}\} \\ &= (A_1 \cup A_2 \cup A_5) \cup (A_3 \cap A_4) \end{aligned}$$

so

$$A^c = (A_1 \cup A_2 \cup A_5)^c \cap (A_3 \cap A_4)^c .$$

Note that in general

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(C \cap B) \\ &\quad + P(A \cap B \cap C) . \end{aligned}$$

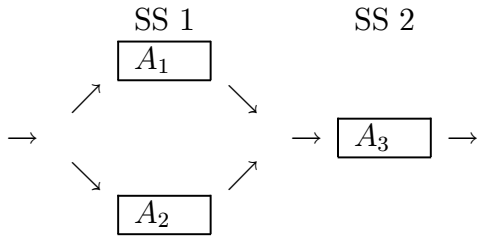
So

$$\begin{aligned} P(A_1 \cup A_2 \cup A_5) &= 3 \times 0.05 - 3 \times 0.05^2 + (0.05)^3 \\ &= 0.142\,6. \end{aligned}$$

Hence

$$\begin{aligned} P(\text{System works}) &= (1 - 0.142\,6) \times \left(1 - (0.1)^2\right) \\ &= 0.848\,8. \end{aligned}$$

12. The system is as follows, in which "SS" stands for "subsystem":



We have

$$P\left(A_i \text{ works}\right) = 0.8$$

for $i = 1, 2, 3$, and so

$$\begin{aligned} P\left(\text{System works}\right) &= P\left(\{\text{SS1 works}\} \cap \{\text{SS2 works}\}\right) \\ & \quad P\left(\text{SS1 works}\right) P\left(\text{SS2 works}\right) \end{aligned}$$

by independence. Now

$$\begin{aligned} P\left(\text{SS1 works}\right) &= P\left(A_1 \text{ works or } A_2 \text{ works}\right) \\ &= P\left(A_1 \text{ works}\right) + P\left(A_2 \text{ works}\right) - P\left(A_1 \text{ and } A_2 \text{ work}\right) \\ &= 0.8 + 0.8 - 0.8 \times 0.8 \\ &= 0.96. \end{aligned}$$

Clearly we have

$$P\left(\text{SS2 works}\right) = P\left(A_2 \text{ works}\right) = 0.8$$

and so

$$\begin{aligned} P\left(\text{System works}\right) &= 0.96 \times 0.8 \\ &= 0.768. \end{aligned}$$

13. (a)

$$P\left(\text{Pass}|M\right) = \frac{24}{40}.$$

(b)

$$P\left(M|\text{Pass}\right) \frac{24}{60}.$$

(c) If independent,

$$\begin{aligned} P(M \text{ and Pass}) &= \frac{24}{100} \\ &= P(M)\,P(\text{Pass}) \\ &= \frac{40}{100} \times \frac{60}{100} \\ &= \frac{6}{25}. \end{aligned}$$

So independence seems to hold here.

14. 1. $S = \{0, 1, 2, \ldots\}$ 2. $S = \{0, 1, 2, \ldots, 365\}$ 3. $S = \{0, 1, 2, \ldots\}$ 4. $S = [0, \infty)$. 5. $S = \{0, 1, 2, \ldots\}$ 6. $S = [0, 100]$.

15.

$$\begin{aligned} A \cup B &= \{\text{error H or M or both}\} \\ A \cap B &= \{\text{error H and M}\} \\ A \cap B^c &= \{\text{error H but not M}\} \\ A^c \cap B^c &= \{\text{error neither H nor M}\} \\ A^c \cup B^c &= \{\text{error not H or not M }\}. \end{aligned}$$

Suppose $P(A) = 0.02$, $P(B) = 0.01$ and $P(A \cup B) = 0.023$, then
(a)

$$\begin{aligned} P(A \cap B) &= P(A) + P(B) - P(A \cup B) \\ &= 0.007. \end{aligned}$$

(b)

$$\begin{aligned} P(A^c \cap B^c) &= 1 - P(A \cup B) \\ &= 1 - 0.023 \\ &= 0.977. \end{aligned}$$

(c)

$$\begin{aligned} P(A \cap B^c) &= P(A) - P(A \cap B) \\ &= 0.02 - 0.007 \\ &= 0.013. \end{aligned}$$

(d)

$$\begin{aligned} P(A|B^c) &= \frac{P(A \cap B^c)}{P(B^c)} \\ &= \frac{0.013}{0.99} \\ &= 0.0131. \end{aligned}$$

(e)

$$\begin{aligned} P(A|B) &= \frac{P(A \cap B)}{P(B)} \\ &= 0.7. \end{aligned}$$

16. In table 3.5, test 2 scores define the columns.
(a) Note that 140 is 9.33% of 1500, whereas 180 is 12%, so presumably we are looking to sift out the top 12% of applicants or thereabouts. Denoting scores in the form {test 1, test 2} we would obviously wish to take those scoring $\{5,5\}$, $\{5,4\}$ and $\{4,5\}$, which amounts to 6% of applicants. Next most favourable are those with a total score of 8, that is, those with $\{4,4\}$, $\{3,5\}$ or $\{5,3\}$, but together these account for a further 8%, tipping us over the desired 12%. There is ambiguity here, since it is unclear how one decides that, say, $\{3,5\}$ is a preferable score to $\{5,3\}$ simply because fewer people achieved the former. The inclusion of $\{5,3\}$ takes the expected percentage to 12%, but appears somewhat unfair on those scoring either $\{4,4\}$ or $\{3,5\}$.

(b) Here the proportion scoring 5 on at least one test is

$$0.10 + 0.05 - 0.02 = 0.13$$

but 13% is too large. Presumably we are expected to exclude those scoring $\{3,5\}$, but again this seems arbitrary.

(c) Note that test 2 is obviously the more difficult.

17. xxx

18. 1. You would probably answer "yes" to all these. 2. 0.58. 3. Sunny and cold days are independent, rainy and cold days are not independent.

19. xxx

20. Recalling that in general

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

applying this twice we observe

$$\begin{aligned} P(A \cap B \cap C) &= P((A \cap B) \cap C) \\ &= P(C|A \cap B)\, P(A \cap B) \\ &= P(C|A \cap B)\, P(B|A)\, P(A) \end{aligned}$$

as required.

21. Letting

$$\begin{aligned} A &= \{\text{Fail midterm 1}\} \\ B &= \{\text{Fail midterm 2}\} \\ C &= \{\text{Fail final exam}\} \end{aligned}$$

then the result of the previous question is useful.

(a) We have

$$\begin{aligned} P(A \cap B) &= P(A)\, P(B|A) \\ &= 0.2 \times 0.6 \\ &= 0.12. \end{aligned}$$

(b) From question 20,

$$\begin{aligned} P(A \cap B \cap C) &= 0.8 \times 0.12 \\ &= 0.096. \end{aligned}$$

22. (a) If we let X denote the lifetime, in hours, and define the events

$$\begin{aligned} A &= \{X > 300\}, \\ B &= \{X > 600\}, \\ C &= \{X > 900\}, \end{aligned}$$

we have

$$\begin{aligned} P(X > 600) &= P(B) \\ &= P(B \cap A) \\ &= P(A)\, P(B|A) \\ &= 0.8 \times 0.6 \\ &= 0.48. \end{aligned}$$

(b) Similarly

$$\begin{aligned} P(X > 900) &= P(C) \\ &= P(C \cap B) \\ &= P(B)\, P(C|B) \\ &= 0.48 \times 0.5 \\ &= 0.24. \end{aligned}$$

23. 2. $\pi = 0.47$.

24. From the information given, we may deduce that

$$\begin{aligned} P(\text{Flaw}) &= 0.005, \\ P(\text{No flaw}) &= 0.995, \\ P(\text{test +ve} \mid \text{Flaw}) &= 0.98, \\ P(\text{test –ve} \mid \text{No flaw}) &= 0.94, \\ P(\text{ test +ve} \mid \text{No flaw}) &= 0.06. \end{aligned}$$

(1) Hence

$$\begin{aligned} P(\text{test +ve}) &= P(\text{test +ve} \mid \text{Flaw})\, P(\text{Flaw}) + P(\text{test –ve} \mid \text{No flaw})\, P(\text{No flaw}) \\ &= 0.98 \times 0.005 + 0.06 \times 0.995 \\ &= 0.0646. \end{aligned}$$

(2) From the above

$$\begin{aligned} P(\text{No flaw} \mid \text{test +ve}) &= \frac{P(\text{test +ve} \mid \text{No Flaw})\, P(\text{No Flaw})}{P(\text{test +ve})} \\ &= \frac{0.06 \times 0.995}{0.0646} \\ &= 0.924. \end{aligned}$$

The test is therefore giving potentially misleading information when a positive result is returned – too often false positives will be observed.

25. Via Bayes' theorem we find:
    1. $P(G|R_1 < R_2 < R_3) = 0.667$, $P(B|R_1 < R_2 < R_3) = 0.333$.

    2. $P(G|R_1 < R_3 < R_2) = 0.294$, $P(B|R_1 < R_3 < R_2) = 0.706$.

    3. $P(G|R_3 < R_1 < R_2) = 0.0667$, $P(B|R_3 < R_1 < R_2) = 0.933$.

    4. For instance, in (1), we would classify the rock as G. The mis-classification errors are

$$\begin{aligned} P(B \cap \{R_1 < R_2 < R_3\}) &= P(R_1 < R_2 < R_3|B)\, P(B) = 0.10 \times 0.75 = 0.075, \\ P(G \cap \{R_1 < R_3 < R_2\}) &= P(R_1 < R_3 < R_2|G)\, P(G) = 0.25 \times 0.25 = 0.0625, \\ P(G \cap \{R_3 < R_1 < R_2\}) &= P(R_3 < R_1 < R_2|G)\, P(G) = 0.15 \times 0.25 = 0.0375 \end{aligned}$$

    Summing the three mis-classification probabilities (as they relate to mutually exclusive events) gives a total mis-classification chance of 0.175.

26. (a) The probability of no errors is

$$0.999^{3500} = 0.0301.$$

    To find the probability of more than one error, note

$$P(> 1 \text{ error}) = 1 - P(0 \text{ errors}) - P(1 \text{ error}).$$

    Now there are 3500 ways a single error can occur, so

$$\begin{aligned} P(1 \text{ error}) &= 3500 \times 0.001 \times 0.999^{3499} \\ &= 0.105\,61. \end{aligned}$$

    Hence

$$\begin{aligned} P(> 1 \text{ error}) &= 1 - 0.0301 - 0.105\,6 \\ &= 0.864\,3. \end{aligned}$$

    (b) Note here an error can occur only when an error occurs at the same point in the string on each transmission. The event that no error occurs can be shown to have probability 0.9965, since if

$$E_{ki} = \{\text{error in transmission } k \text{ in position } i\},$$

for $i = 1, \ldots, 3500$ and $k = 1, 2$, then if E is the event an error occurs (i.e., the error checking fails to detect an error),

$$\begin{aligned} P(E^c) &= P\left(\bigcap_{i=1}^{3500} (E_{1i} \cap E_{2i})^c\right) \\ &= \prod_{i=1}^{3500} (1 - P(E_{1i})\, P(E_{2i})) \\ &= \left(1 - (0.001)^2\right)^{3500}. \end{aligned}$$

27. (a)

$$\begin{aligned} P(A^c \cap B^c) &= 1 - P(A \cup B) \\ &= 1 - (P(A) + P(B) - P(A)\, P(B)) \\ &= 1 - (P(A) + P(B) - (1 - P(A^c))\,(1 - P(B^c))) \\ &= P(A^c) - P(B) + 1 - P(B^c) - P(A^c) + P(A^c)\, P(B^c) \\ &= P(A^c)\, P(B^c). \end{aligned}$$

(b)

$$\begin{aligned} P((A \cap C) \cup (B \cap C)) &= P(A \cap C) + P(B \cap C) - P(A \cap B \cap C) \\ &= P(A)\, P(C) + P(B)\, P(C) - P(A)\, P(B)\, P(C) \\ &= P(C)\,(P(A) + P(B) - P(A)\, P(B)) \\ &= P(C)\, P(A \cup B). \end{aligned}$$

28. xxx

29. (1) Here

$$P(B_i \cup C_j) = P(B_i) + P(C_j) - P(B_i \cap C_j)$$

for $i, j = 1, 2, 3$. For instance

$$\begin{aligned} P(B_1 \cup C_1) &= 0.2 + 0.2 - 0.1 \\ &= 0.3, \end{aligned}$$

$$\begin{aligned} P(B_1 \cup C_2) &= 0.2 + 0.6 - 0.46 \\ &= 0.72, \\ P(B_2 \cup C_2) &= 0.6 + 0.6 - 0.46 \\ &= 0.74, \\ P(B_3 \cup C_3) &= 0.2 + 0.2 - 0.11 \\ &= 0.29. \end{aligned}$$

(2) The event A in effect is the event that an individual falls in the same class for both GPA and salary. As the three events, $B_i \cap C_i$ are mutually exclusive, for $i = 1, 2, 3$, we have

$$P(A) = 0.1 + 0.46 + 0.11 = 0.67.$$

(3) Salary and GPA are clearly not independent. Recall that events A and B are independent if

$$P(A \cap B) = P(A) P(B).$$

Here, for instance,

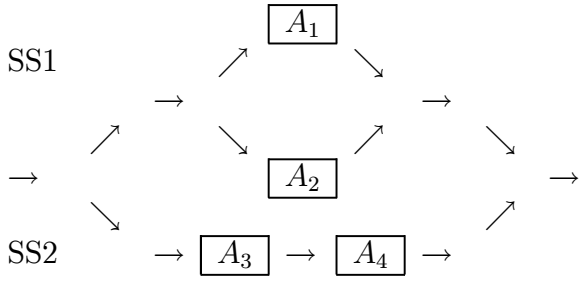
$$\begin{aligned} P(B_1 \cap C_1) &= 0.1 \neq 0.2 \times 0.2, \\ P(B_2 \cap C_1) &= 0.07 \neq 0.6 \times 0.2. \end{aligned}$$

(3) We construct the "joint distribution" by multiplying the marginal probabilities together as below:

		Salary			
		L	M	H	
GPA	L	0.04	0.12	0.04	0.20
	M	0.12	0.36	0.12	0.60
	H	0.04	0.12	0.04	0.20
		0.2	0.6	0.2	1.00

30. (a) With the two subsystems, SS1 and SS2, in parallel the system is as

below:



We know

$$P(A_i) := P(A_i \text{ works}) = 0.85$$

for $i = 1, 2, 3, 4$, and so

$$\begin{aligned} P(\text{System work}) &= P(\text{SS1 works or SS2 works or both}) \\ &= P(\text{SS1 works}) + P(\text{SS2 works}) - P(\text{both work}). \end{aligned}$$

Now

$$\begin{aligned} P(\text{SS1 works}) &= P(A_1 \text{ or } A_2 \text{ or both work}) \\ &= P(A_1) + P(A_2) - P(A_1)\,P(A_2) \\ &= 0.85 + 0.85 - 0.85^2 \\ &= 0.9775. \end{aligned}$$

Further,

$$\begin{aligned} P(\text{SS2 works}) &= P(A_3 \cap A_4) \\ &= 0.85^2 \\ &= 0.7225, \end{aligned}$$

and we deduce

$$\begin{aligned} P(\text{System works}) &= 0.9775 + 0.7225 - 0.9775 \times 0.7225 \\ &= 0.9938. \end{aligned}$$

(b) With SS1 and SS2 in series,

$$\begin{aligned} P\,(\text{System works}) &= P\,(\text{SS1 and SS2 work}) \\ &= P\,(\text{SS1 works})\,P\,(\text{SS2 works}) \\ &= 0.9775 \times 0.7225 \\ &= 0.706\,2. \end{aligned}$$

31. $P\,(\text{System works}) = 0.948$.

BD